

A representative of the unique irreducible class

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.IsSingletonIrreducibleModel`

def

Defines when one nonzero irreducible $*$ -representation represents the sole unitary-equivalence class of such representations.

Statement

Let A be a complex C^* -algebra, with no unit assumed, and let $\pi : A \rightarrow \mathcal{B}(H)$ be a $*$ -representation. The predicate below says that π is nonzero irreducible and that every other nonzero irreducible $*$ -representation of A is unitarily equivalent to π .

Definition

Require first that π act nontrivially and have no proper nonzero closed reducing subspace. Then, for every complex Hilbert space K and every nonzero irreducible $*$ -representation $\rho : A \rightarrow \mathcal{B}(K)$, require a unitary operator $U : H \rightarrow K$ such that

$$U\pi(a) = \rho(a)U \quad (a \in A).$$

Assumptions

No separability or faithfulness condition is part of this definition. The algebra may be unital, but the definition does not require a unit or a unit-preservation equation.

Conclusion

The output is a property of the specified π : it represents the unique unitary-equivalence class of nonzero irreducible $*$ -representations of A .

Faithfulness is not part of this definition. Later theorems impose hypotheses to deduce it. “Non-unital” permits an algebra with a unit; it does not assert that a unit fails to exist.

Main citations

- Exact definition and context
- Nonzero irreducibility
- Unitary equivalence

Lean source signature (exact)

The complete declaration below is a separate exact source excerpt; the original header record is retained with the manuscript.

```
-- A displayed nonzero irreducible representation representing every
nonzero irreducible representation of a genuinely nonunital C-star algebra.
Faithfulness is deliberately not part of this predicate. -/
def IsSingletonIrreducibleModel
  (pi : NonUnitalCStarRepresentation A H) : Prop :=
  pi.IsIrreducible ∧
  ∀ (K : Type w) [NormedAddCommGroup K] [InnerProductSpace ℂ K]
    [CompleteSpace K] (rho : NonUnitalCStarRepresentation A K),
    rho.IsIrreducible → pi.UnitaryEquivalent rho
```

In the source	Mathematical meaning
(pi : NonUnitalCStarRepresentation A H)	The chosen representation $\pi : A \rightarrow \mathcal{B}(H)$, without a unit-preservation requirement.
: Prop	The output is a condition on this π .
pi.IsIrreducible ∧	First require that some $a \in A$ acts by a nonzero operator and that the only closed reducing subspaces are $0, H$; the comparison condition must hold as well.
∀ (K : Type w)	For every comparison space K in its own size universe.
[NormedAddCommGroup K] [InnerProductSpace ℂ K] [CompleteSpace K]	For every complete complex Hilbert-space structure on K ; separability is not added.
(rho : NonUnitalCStarRepresentation A K)	For each representation ρ of the same A on that K .
rho.IsIrreducible → pi.UnitaryEquivalent rho	If ρ is nonzero irreducible, there is a unitary $U : H \rightarrow K$ intertwining every $\pi(a)$ and $\rho(a)$ as in the displayed equation.

Exact source and proof.

Earlier published Card and PDF

▼ Exact content identity

Declaration: MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.IsSingletonIrreducibleModel
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